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Machine Learning and its Applications

Today's goals

Module 1
Machine Learning Fundamentals I: Basic Concepts

Module 2
Machine Learning Fundamentals II: Model Training and Evaluation

Module 3
Data Preparation

Module 4
Support Vector Machines

Module 10 Introduction

Introduction to Module 10's Study
Video • 1 min

Why Support Vector Machines?
Reading • 5 min

Module 10-1 Support Vector Machine Fundamentals

Support Vector Machine Fundamentals: Concepts
Video • 11 min

Support Vector Machine Fundamentals: Types of SVM
Video • 4 min

Materials on Support Vector Machine Fundamentals
Reading • 10 min

Module 10-2 Support Vector Machines

Support Vector Machines: Linear SVM of Hard Margin Classifier
Video • 7 min

Support Vector Machines: Linear SVM of Soft Margin Classifier
Video • 11 min

Support Vector Machines: Non-Linear SVM
Video • 12 min

Support Vector Machines: Multi-Class SVM
Video • 8 min

Materials on Support Vector Machines
Reading • 10 min

Module 10-3 Support Vector Machine Implementation and Case Studies

Support Vector Machine Implementation: MATLAB Implementation of SVM
Video • 4 min

Support Vector Machine Implementation: Iris Flower Example
Video • 5 min

Support Vector Machine Implementation: 2D Point Classification Example
Video • 35 sec

Case Study 1: Banknote Classification (Linear SVM)
Video • 4 min

Case Study 2: Raisin Classification (Non-Linear SVM)
Video • 7 min

Materials on Support Vector Machine Implementation and Case Studies
Reading • 10 min

Support Vector Machine Learning Coach
Dialogue • 5 min

Module 10 Quiz

Module 10 Quiz
Practice Assignment • Grade: 100%

Module 10 Assignments

Module 10 Assignment 1
Graded App Item • 30 min

Module 10 Assignment 2
Graded App Item • 30 min

CSV file for Module 10 Assignment
Reading • 1 min

Module 10 Recap
Reading • 5 min

Module 5
Artificial Neural Networks

Module 10 Recap

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Over this module, you studied **Support Vector Machines (SVMs)** from fundamental concepts to practical MATLAB implementations, including linear and nonlinear decision boundaries, margin maximization, kernel methods, and real-world case studies.

Before moving on to **Artificial Neural Networks**, reflect on the following:

- What are the key strengths of SVMs that make them effective for many classification problems, especially with limited or high-dimensional data?
- In practice, what limitations did you observe or anticipate when working with SVMs (e.g., kernel selection, scalability, interpretability, or hyperparameter tuning)?
- Under what circumstances might SVMs no longer be the most suitable choice, motivating the use of more flexible models such as neural networks?
- Based on your experience so far, what capabilities do you expect Artificial Neural Networks to offer that extend beyond what SVMs can achieve?

Use concepts from this module—such as **margins, kernels, generalization, and model evaluation**—to support your discussion. This reflection will help you see SVMs not as isolated tools, but as part of a broader machine learning landscape that naturally leads into neural network–based models.

Go to next item ✓ Completed

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